Data Visualization 1

Jameson Roth

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# ============================================
# Program: base_pop_data_vis.R
# Author: Jameson Roth
# Date: 2026 10 27
# Description: Data visualisations and summary stats of base population (2011 zips)
# ============================================

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------------------------------------------
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------------------------------------------------
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(rprojroot)

setwd(find_root(is_git_root))

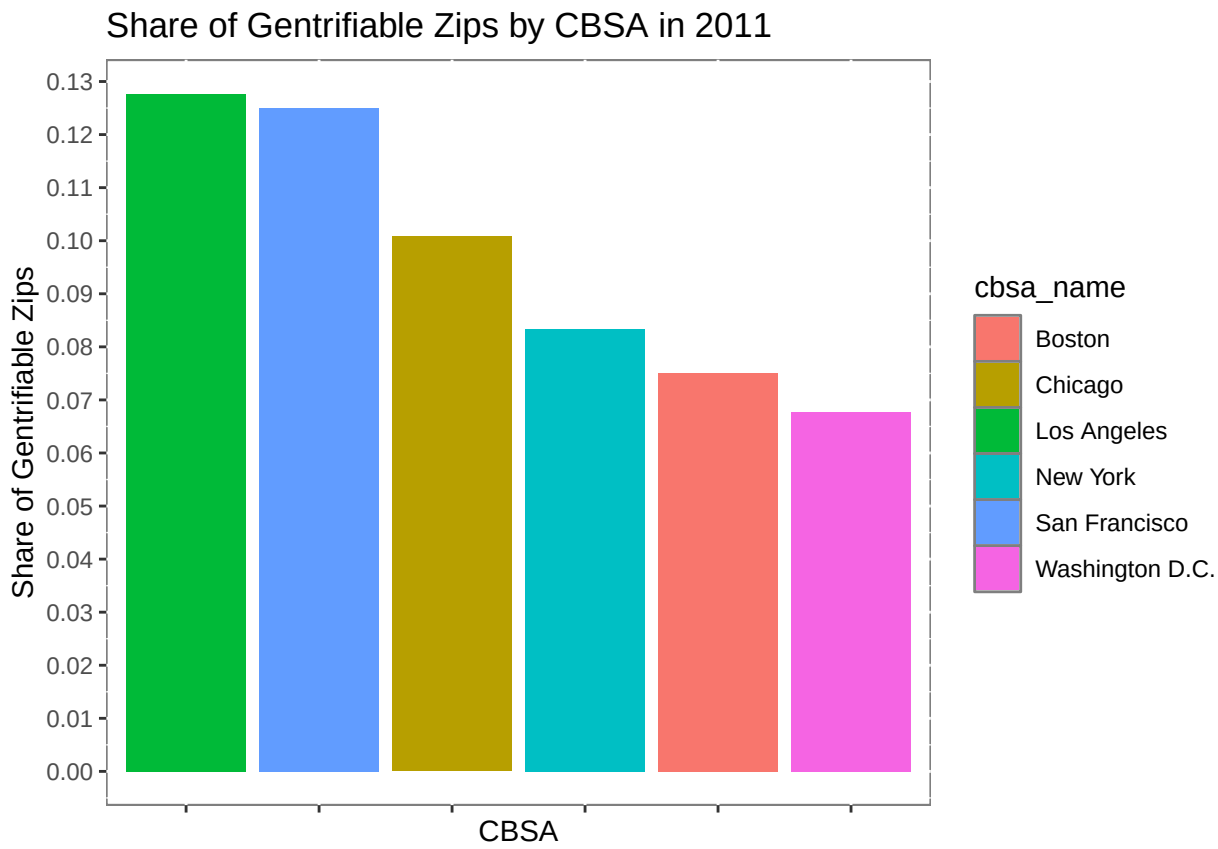
pop <- read_csv("0_data/1_clean/base_population.csv") |>
    group_by(cbsa) |>
    mutate(
        count_gentrifiable = sum(gentrifiable, na.rm = TRUE),
        total_zips = n(),
        share_gentrifiable = count_gentrifiable / total_zips,
        cbsa = as.character(cbsa),
        cbsa_name = case_when(
            cbsa == "14460" ~ "Boston",
            cbsa == "47900" ~ "Washington D.C.",
            cbsa == "31080" ~ "Los Angeles",
            cbsa == "41860" ~ "San Francisco",
            cbsa == "35620" ~ "New York",
            cbsa == "16980" ~ "Chicago",
        )
    ) |>
    ungroup() |>
    select(cbsa, count_gentrifiable, share_gentrifiable, cbsa_name) |>
    unique()
```

```
## Rows: 2346 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): zipcode
## dbl (8): statefips, year, cbsa, zip_median, cbsa_40pct, annual_avg_zhvi, med...
## lgl (1): gentrifiable
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
summary(pop)
```

```
##         cbsa   count_gentrifiable share_gentrifiable     cbsa_name
##  Length   :6   Min.   :20.00      Min.   :0.06774    Length   : 6
##  N.unique :6   1st Qu.:21.25      1st Qu.:0.07715    N.unique : 6
##  N.blank  :0   Median :30.50      Median :0.09205    N.blank  : 0
##  Min.nchar:5   Mean   :36.33      Mean   :0.09661    Min.nchar: 6
##  Max.nchar:5   3rd Qu.:45.00      3rd Qu.:0.11894    Max.nchar:15
##                Max.   :69.00      Max.   :0.12772
```

```
# Share gentrifiable by CBSA
chart <- ggplot(
    pop,
    aes(
        x = reorder(cbsa_name, -share_gentrifiable),
        y = share_gentrifiable,
        fill = cbsa_name
    )
) +
    geom_bar(stat = "identity") +
    xlab("CBSA") +
    ylab("Share of Gentrifiable Zips") +
    scale_y_continuous(breaks = seq(0, 0.5, 0.01)) +
    ggtitle("Share of Gentrifiable Zips by CBSA in 2011") +
    theme(
        panel.background = element_rect(
            fill = "white",
            colour = "grey50"
        ),
        axis.text.x = element_blank()
    )
show(chart)
```


Share of Gentrifiable Zips by CBSA in 2011
Share of Gentrifiable Zips
0.13
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
CBSA
cbsa_name
Boston
Chicago
Los Angeles
New York
San Francisco
Washington D.C.